

Retrieval-Augmented Generation (RAG) Architecture

Concept of RAG Retrieval-Augmented Generation (RAG) enhances LLMs by retrieving relevant documents from external sources and injecting them into the LLM context before generation, reducing hallucinations and enabling queries on private data.

RAG Life Cycle 1. Ingestion: Documents are loaded, parsed, split into smaller chunks, and embedded. 2. Indexing: Chunks are stored in a Vector DB index. 3. Retrieval: The user query is embedded, and the database returns the top-K similar chunks. 4. Generation: Chunks and query are assembled into a prompt. The LLM generates the grounded response.

Advanced RAG Advanced strategies include query rewriting, hybrid search (combining dense vector search with BM25 keyword matching), and reranking (using cross-encoder models to reorder top retrieved documents for better relevance).